

# Introduction to AI Safety, Ethics, and Society — Ch2 AI Fundamentals Quiz

**1. Which of the following best matches the course's working definition of “artificial intelligence”?**

- A) A single universally accepted technical standard
- B) Computer systems that perform tasks typically associated with intelligence, such as problem solving, decision making, and forecasting
- C) Any computer program that processes data
- D) A branch of robotics exclusively

ANS:

**3. Which development is credited in the timeline as the “first neural network”?**

- A) The Perceptron
- B) The Transformer
- C) ImageNet
- D) ChatGPT

ANS:

**4. Which type of AI is defined as “capable of performing specific tasks only”?**

- A) Narrow AI
- B) Artificial General Intelligence (AGI)
- C) Human-level AI (HLAI)
- D) Superintelligence (ASI)

ANS:

**5. Which type of AI is defined as “capable of performing many cognitive tasks across multiple domains, at human or superhuman levels”?**

- A) Narrow AI
- B) Artificial General Intelligence (AGI)
- C) Transformative AI (TAI)
- D) Superintelligence (ASI)

ANS:

**6. Which type of AI is defined as “capable of outclassing human performance on virtually all intellectual tasks”?**

- A) Narrow AI
- B) Artificial General Intelligence (AGI)
- C) Human-level AI (HLAI)
- D) Superintelligence (ASI)

ANS:

**7. According to the slides, what is “machine learning”?**

- A) Writing explicit rule-based programs for every task
- B) Developing computer systems that can learn directly from data without following explicit pre-set instructions

- C) A synonym for artificial general intelligence
- D) A hardware specification for GPUs

ANS:

**8. How do the slides define an “algorithm”?**

- A) A dataset used to train models
- B) A procedure (recipe) for accomplishing a task
- C) A type of neural network layer
- D) A hardware accelerator

ANS:

**9. What is “training data,” as defined in the slides?**

- A) The output produced by a deployed model
- B) The data from which ML models learn
- C) A model's final accuracy score
- D) A type of activation function

ANS:

**10. Which model access category is described as “accessible through an interface with usage restrictions, allowing users to query the model without viewing or modifying the underlying model”?**

- A) Closed models
- B) API models
- C) Open models
- D) Federated models

ANS:

**11. Which model access category is “fully accessible and modifiable, often part of open-source projects”?**

- A) Closed models
- B) API models
- C) Open models
- D) Sandboxed models

ANS:

**12. Which computational resource is characterized in the slides by “parallel processing” and is a primary example of compute used in ML?**

- A) CPU
- B) GPU
- C) Hard disk drive
- D) Network router

ANS:

**13. Which ML task is defined as “predicting categories or ‘classes’ using features of an input data point”?**

- A) Regression
- B) Classification

- C) Sequence modeling
- D) Anomaly detection

ANS:

**14. Which ML task is illustrated in the slides using the example of “black swan detection”?**

- A) Classification
- B) Regression
- C) Anomaly detection
- D) Generative modeling

ANS:

**15. Which evaluation metric is described in the slides as “the purity of the flagged examples”?**

- A) Accuracy
- B) Precision
- C) Recall
- D) Latency

ANS:

**16. Which evaluation metric is described as “the proportion of relevant examples that were flagged”?**

- A) Accuracy
- B) Precision
- C) Recall
- D) Transparency

ANS:

**17. Which type of machine learning is defined as “learning from labeled data”?**

- A) Unsupervised learning
- B) Supervised learning
- C) Reinforcement learning
- D) Self-play learning

ANS:

**18. Which type of machine learning is also referred to as “self-supervised learning,” since the model may generate its own supervisory signals from patterns in the data?**

- A) Supervised learning
- B) Unsupervised learning
- C) Reinforcement learning
- D) Active learning

ANS:

**19. Which type of machine learning involves an agent making decisions and improving its performance based on responses from its environment?**

- A) Supervised learning
- B) Unsupervised learning
- C) Reinforcement learning
- D) Transfer learning

ANS:

**20. According to the slides, what do deep learning (DL) models use to extract useful patterns from large quantities of data?**

- A) Decision trees
- B) Neural networks: layers of interconnected nodes inspired by biological neurons
- C) Rule-based expert systems
- D) Manually engineered lookup tables

ANS:

**21. What is a “multi-layer perceptron (MLP),” according to the slides?**

- A) A type of activation function
- B) A foundational neural network architecture with input, hidden, and output layers
- C) A method for evaluating model accuracy
- D) A dataset labeling technique

ANS:

**22. What is the main purpose of “residual connections” in deep learning, per the slides?**

- A) To increase the model's parameter count
- B) To provide pathways for information to bypass network layers, helping preserve original content and learn hierarchical features
- C) To reduce the size of training datasets
- D) To replace activation functions entirely

ANS:

**23. How do the slides describe “self-attention”?**

- A) A method for compressing model weights
- B) A mechanism that encodes relationships between elements in a sequence by having each element “attend” to every other element
- C) A type of loss function used only in classification
- D) A hardware optimization for GPUs

ANS:

**24. According to the slides, what is a “Transformer”?**

- A) A type of activation function
- B) A deep learning architecture used in LLMs that combines self-attention and MLPs with techniques such as residual connections
- C) A dataset labeling tool
- D) A type of reinforcement learning algorithm

ANS:

**25. What is the main takeaway from “The Bitter Lesson,” as described in the slides?**

- A) Expert knowledge always outperforms scale
- B) Increasing the scale of computation and data tends to beat complex, hand-engineered research approaches in the long run
- C) Deep learning models cannot scale well
- D) Scaling laws only apply to image classifiers

ANS: